

HAZEM MARRAKCHI

Software Engineer | Full Stack Developer

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PROFESSIONAL PROFILE

EUR-ACE accredited Software Engineering graduate (Distinction, 18.5/20) building modern full-stack applications and microservices, from architecture modeling to end-to-end testing. TypeScript ecosystem: React on the frontend, Node.js and NestJS on the backend, with solid professional experience in Java Spring Boot and Python FastAPI. Differentiator: turned the European e-invoicing regulation EN 16931 into two real, deployed products — Facturia and eInvoice Studio. Autonomous and rigorous; treat quality as part of development: GitHub Actions CI/CD, Playwright and Vitest automated tests, Docker containerization.

SKILLS

Langages : Java, TypeScript, JavaScript, Python, PHP, SQL, C++, HTML5/CSS3

Frontend : React 19, Angular, Three.js, Tailwind CSS v4, Bootstrap

Backend : Node.js, NestJS, Express.js, Spring Boot, FastAPI, REST APIs, WebSockets, .NET

BDD & ORM : MongoDB, MySQL, Oracle, SQLite, Drizzle ORM, Zod

IA & ML : Scikit-learn, Deep Learning

Cloud & DevOps : Docker, Microsoft Azure, Linux, GitHub Actions, CI/CD, OpenAPI/Swagger

Tests : Cypress E2E, Playwright E2E, Vitest, axe-core (a11y)

Outils & Méthodes : Git, GitHub, Postman, Agile Scrum, UML, n8n

PROJECTS

INVOICE STUDIO — *Public E-Invoice Validation Tool*

Open source project

- Public sandbox validating UBL 2.1 / Peppol BIS invoices against EN 16931 rules: instant, clear, rule-by-rule diagnostics with anomaly explanation and suggested fix.
- Visual exploration of invoice structure: parties, totals, line-by-line breakdown, readable renderings.
- Namespace-agnostic parser and pure-TypeScript rule engine — full privacy (no data leaves the browser).
- Vitest unit tests blocking deployment in CI.

Stack: React 19, TypeScript, Vite, Tailwind CSS v4, Vitest

[Demo](#) | [Code](#)

FACTURIA — *EU-Compliant E-Invoicing SaaS Platform*

Personal project

- Generate UBL 2.1 invoices compliant with Peppol BIS Billing 3.0 and XRechnung, auto-validated against EN 16931 business rules.
- Full invoice lifecycle (draft -> sent -> accepted -> rejected -> paid), per-organization sequential numbering, client directory.
- Multi-organization architecture with role-based access control, immutable RGPD audit trail with export and data deletion.
- NestJS REST API documented with OpenAPI/Swagger; React frontend deployed on GitHub Pages consuming a live API hosted on Render.
- Quality: 8 Playwright end-to-end tests run against a real server in CI, GitHub Actions pipeline, Docker images.

Stack: NestJS, React 19, TypeScript, Drizzle ORM, SQLite, Zod, Tailwind CSS v4, Docker

[Demo](#) | [Code](#)

SANA TRAVEL — *Trilingual Travel Agency Platform*

Production

- Trilingual (FR/EN/AR+RTL) platform in production: travel catalogue, cascading hotel search, real-time flights (Amadeus API), trip builder, AI chatbot.
- Newsletter, JWT authentication, NestJS REST API + MongoDB Atlas.

Stack: React 19, TypeScript, Vite, NestJS, MongoDB Atlas, Tailwind CSS v4, JWT, Amadeus API

[Demo](#) | [Code](#)

Other projects

- **LENDSWIFT** — Credit Application Platform : Premium 8-step credit application wizard (Indian market): real-time Zod validation, conditional co-applicant logic, KYC verification (PAN/Aadhaar with Verhoeff checksum).
- **MEDAI PRO** — Intelligent Hospital Platform : Intelligent hospital platform with AI predictions: real-time coordination between hospital departments via WebSockets.

- **CINEMA BOOKING & CARPOOLING** — Academic projects : Cinema ticket booking: session management, seat selection and Spring Boot REST backend on MySQL.

LENDSWIFT : React 19, TypeScript, Vite 8, Tailwind CSS v4, React Hook Form, Zod 4, Cypress, Playwright, Vitest | **MEDAI PRO** : Angular 17, Node.js, Express.js, FastAPI, Python, Scikit-learn, MongoDB, WebSockets, JWT | **CINEMA BOOKING & CARPOOLING** : Angular, Spring Boot, PHP, JavaScript, Bootstrap, MySQL

PROFESSIONAL EXPERIENCE

Software Engineering Intern — *Digilife, Gabès* (Feb 2026 – Jun 2026)

- Designed and developed an intelligent hospital platform integrating AI predictions (Angular, Node.js, Express.js, FastAPI, MongoDB).
- Contributed to software architecture design and JWT-secured REST API development.

Software Engineering Intern — *Telecom Operator, Gabès* (Jun 2025 – Jul 2025)

- Developed web and mobile applications, showcase sites and internal tools.
- Deployed and supported maintenance of IT and application infrastructures.

Software Engineering Intern — *Telecom Operator, Gabès* (Jun 2024 – Jul 2024)

- User support, maintenance of IT park, network and messaging.
- Created queries and updated databases; proposed automation solutions.

IT Intern — *STEG, Gabès* (Jun 2022 – Jul 2022)

- Maintenance and IT support activities; assistance in managing IT environments.

Freelance Web Developer — *Sana Travel Agency, Gabès* (Apr 2018 – Oct 2020)

- Built and maintained the showcase site and online booking platform (front-end and back-end).
 - Integrated airline, hotel and payment provider APIs.
 - SEO and performance optimization; security and updates.
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EDUCATION

National Engineering Degree in Software Engineering — 2023–2026

ESSAT Gabès, Tunisia — EUR-ACE® accredited (ASIIN e.V.) — Distinction — 18.5/20

Bachelor in Management Computing — 2020–2023

ESSAT Gabès, Tunisia

Baccalaureate in Economics & Management — 2017–2018

Gabès, Tunisia

CERTIFICATIONS & LANGUAGES

- Full Stack & Front End Development — LinkedIn Learning
- Deep Learning Foundation
- Front End Developer — Multi Step Loan Application (ZeTheta)

Arabic : Native • **French** : Professional • **English** : Professional • **German** : Beginner (A1)

Permis : Driving Licence B (car)